MICHAL BOREK

AI Engineer · Applied AI, Computer Vision & Machine Learning
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PROFESSIONAL SUMMARY

AI Engineer with 3+ years of experience developing and deploying AI systems, specializing in computer vision, foundation models, and applied machine learning. Experienced in building end-to-end AI applications, from model development and optimization to software architecture and deployment. Combines published research with hands-on engineering experience delivering reliable AI solutions for real-world users.

TECHNICAL SKILLS

AI & Machine Learning: Python, PyTorch, Hugging Face, Computer Vision, Vision Transformers, Multimodal AI, LLMs, VLMs, LoRA/PEFT, Model Fine-Tuning, RAG
Software & AI Systems: ONNX, FastAPI, PySide6, REST APIs, Docker, Git, CI/CD, GitHub Actions, pytest, PostgreSQL, Qdrant, Ollama
Cloud & Infrastructure: AWS (SageMaker, S3, Lambda, API Gateway, ECR), CUDA, SLURM, Distributed GPU Training, PyTorch DDP

PROFESSIONAL EXPERIENCE

Machine Learning Researcher | Institute for Quantitative Health Science & Engineering, MSU *June 2026 – Present*

- Led **end-to-end development and September 2026 beta deployment** of an offline AI-powered medical imaging application for radiologists, owning system architecture, model integration, software development, and user feedback.
- Engineered a **Windows desktop application** using Python, PySide6, and ONNX, implementing automated inference validation, CI/CD, installer packaging, and **approximately 1,860 automated tests**.
- Fine-tuned and evaluated vision foundation models, improving chest X-ray classification accuracy from **75% to 85%** and ROC-AUC from **0.80 to 0.96**; adapted MedGemma 1.5 4B with LoRA, increasing accuracy from **82% to 90%**.
- Collaborated with **CDC/NIOSH researchers and radiologists** to translate domain requirements into reliable AI solutions and evaluate model performance in real-world workflows.

Deep Learning Research Assistant | Institute for Quantitative Health Science & Engineering, MSU *November 2023 – May 2026*

- Developed **computer vision pipelines** for medical image classification, including preprocessing, deep learning model training, performance evaluation, and explainability.
- Optimized large-scale GPU training using **PyTorch DDP, CUDA, and SLURM**, reducing training time from **approximately 8 hours to 3 hours**.
- Developed **cloud-based inference workflows** using AWS SageMaker, S3, Lambda, API Gateway, and CloudWatch.

AI Engineer Intern — HPC / ML Tooling | Institute for Cyber-Enabled Research, MSU *June 2023 – May 2026*

- Developed and deployed a **local AI agent** for HPC job validation, combining LLM-based analysis with deterministic checks, Flask APIs, and cluster infrastructure.
- Built a **machine learning resource-recommendation system** using historical computing-job data to estimate CPU, GPU, memory, and execution-time requirements.

SELECTED PROJECT

Legal Document Intelligence System — *Generative AI, RAG, Python, FastAPI*

- Developed a local **AI-powered document intelligence application** using LLMs, hybrid retrieval with PostgreSQL and Qdrant, and citation-grounded answer generation.
- Implemented **structured outputs, source validation**, audit logging, automated testing, and reliability controls for document question answering.

EDUCATION

Michigan State University — College of Engineering
B.S. Computational Data Science · Minor in Mathematics · GPA: 3.80/4.00 · Honors College · Dean's List

PUBLICATIONS

Co-author, *"Ordinal classification framework for multiclass grading of pneumoconiosis,"* Proc. SPIE 13407, Medical Imaging 2025, 134072Q (2025). DOI: 10.1117/12.3046353
Co-author, *"Pneumoconiosis screening and classification using deep learning models,"* Occupational & Environmental Medicine 2026;83(5):276–282. DOI: 10.1136/oemed-2026-111040